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Semantic Enrichment of Taxonomy for BI Applications using Multifaceted data sources through NLP techniques

Muhammad Arslan[a]* and Christophe Cruz[a]

[a]*Laboratoire d'informatique de Bourgogne (LIB), 9 Avenue Alain Savary, Dijon 21000, France*

Abstract

Taxonomies are crucial for executing Business Intelligence (BI) applications by preventing users from being overwhelmed with information. The business applications require knowledge of the key concepts and their organization to perform the classification of news articles. To ensure and maintain reliable information classification quality, it is crucial to keep the same definitions and organization of these concepts. This indicates the major significance of BI taxonomies in organizations. However, their development in business information systems follows an ad hoc process in most cases. Compared to many other domains, e.g. environmental and life sciences research, no mature and updated BI taxonomies are available in the literature. Existing studies cover BI taxonomies, but these are excessively generic and domain-specific. As a result, the BI domain suffers from many immature, incorrect, and incomplete notions of concepts. New BI-related concepts emerge rapidly, making it essential to include them in existing taxonomies during the enrichment process. The contribution of our research is the exploration of the possibilities of taxonomy enrichment using existing datasets. The expansion of the existing business taxonomy using multifaceted data sources to capture new concepts comprising 1) lexical datasets, 2) pre-trained word embeddings, 3) linked open data vocabulary, and 4) corpus-based relevant thematic extraction of features from news articles using Natural Language Processing (NLP) techniques. The highest semantic enrichment rate of a taxonomy got on a combination of these 4 methods. Eventually, enriched business taxonomy will contribute to the improved classification of news articles.

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Keywords: Artificial Intelligence (AI); Business Intelligence (BI); Natural Language Processing (NLP), Lexical Datasets; Word Embeddings

* Corresponding author. Tel.: (33) 0628430801
E-mail address: muhammad.arslan@u-bourgogne.fr

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1. Introduction

Pre-classification of news articles (see Fig. 1) based on business user feedback plays a significant role in averting users from being overwhelmed with information [1]. Manual structuring and classifying news articles are time-consuming, which makes it essential to explore possibilities to automate this information organization process. This could be achieved using state-of-the-art automatic text classification methods [1]. In the existing literature, the classification of news articles is realized using both taxonomy-based and machine learning techniques through textual features extraction from news articles by NLP methods [1]. The focus of this article falls into the first category of classification.

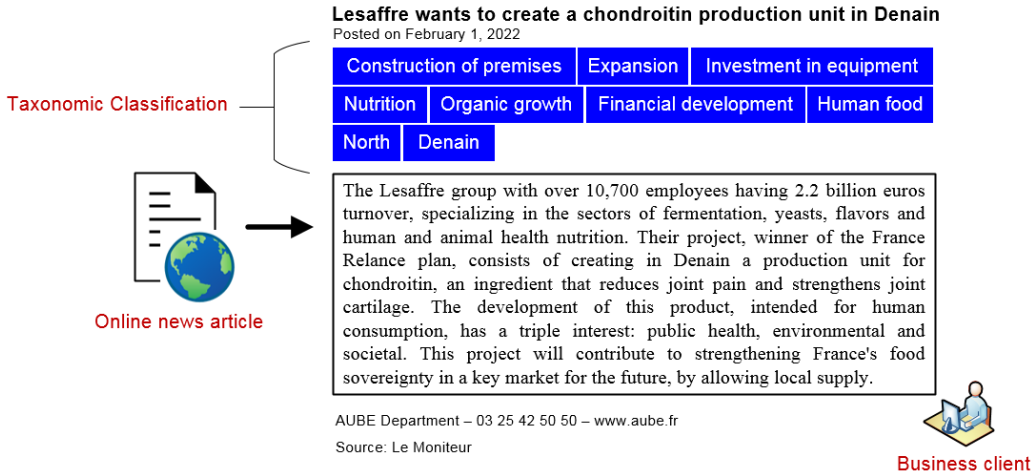

Fig. 1. Classification of a news article for a business client

Taxonomies constitute a valuable resource for knowledge-based enterprises [1]. A taxonomy is a tree-based structure where nodes (concepts) are linked with the hypernymy relation and organized concepts into a hierarchical structure [2]. They are used for searching, classifying, organizing information, and many other tasks. The taxonomies are usually constructed manually, although many techniques exist in the literature to support automatic taxonomy construction [2]. Automatic construction of taxonomies is more error-prone than manually creating taxonomies. But constructing large taxonomies manually is time-consuming and expensive in a dynamic environment because of the emergence of new concepts resulting from the news on emerging topics. Once taxonomies are constructed, a large amount of engineering is required to enrich them with new domain concepts. Semantic enrichment is augmenting a concept hierarchy (i.e. a taxonomy) with novel or unfamiliar contextually-similar concepts at an appropriate location (avoiding too general or too specific concepts) [3, 4]. The enrichment of existing taxonomies allows business users to refine their search queries [3] by choosing more specific or more general updated concepts when the previous queries are returning too many or too few answers or answers are not following the latest business concepts [1 – 3]. Such enrichment of existing business taxonomy with new concepts to better capture the dynamic (i.e. changing) world news presents two-fold challenges: 1) identification of existing BI-specific datasets and vocabularies to enrich taxonomies, and 2) inserting new concepts into the current taxonomy, respecting the actual organization of business concepts.

Our approach to taxonomy enrichment is based on exploiting multifaceted data sources to capture new concepts encompassing; 1) real-world lexical taxonomies including WordNet (i.e. a network of synonym sets representing lexical concepts) and Wiktionary (i.e. a web-based open project with terms having words, phrases, etc.). 2) Linked open data vocabulary such as AGROVOC (i.e. largest and up-to-date linked Open data) to incorporate new concepts which are not present in WordNet and Wiktionary. In addition, pre-trained word embedding models, such as Sense2Vec and Global Vectors for Word Representation (GloVe) are used to retrieve similar taxonomy concepts. To complement this enrichment process to include the most updated concepts which are relevant to the business taxonomy, a corpus-based thematic features extraction technique is used. During the insertion of new concepts into a

business taxonomy, the identified new concepts, as well as taxonomy concepts, are converted into vector representations, and later cosine similarity is used to identify semantic similarities between them. Based on their similarity score threshold, the concepts are filtered for insertion into the business taxonomy.

2. Background

Lexical datasets such as WordNet have been an important resource in NLP and have remained in widespread use over the past decades [5]. It offers a broad range of applications such as Word Sense Disambiguation, Information Retrieval, and semantic similarity measurement [5]. The amount of the lexical coverage of WordNet's vocabulary determines the performance of WordNet-based techniques. Though, the maintenance of WordNet is labor-intensive, as it requires considerable time and effort [5]. Thus, WordNets are not updated frequently and cannot identify many domain-specific lexicons that came into existence recently. In the literature [6 - 8] several methods are present to address the limitation of word coverage of WordNet, often inferring new word senses from domain-specific vocabulary or combining different taxonomies and adding to the WordNet concepts' organization. However, either these methods are tested on small datasets or integration accuracy between the word senses is not precisely determined [5]. Hence, word senses are not well integrated into the taxonomy. The accessibility of large-scale datasets (e.g. Wiktionary) has inspired new approaches to use definitions of concepts for expanding WordNet vocabulary with new concepts [5].

In the literature, the taxonomy enrichment is achieved by; 1) leveraging knowledge transfer techniques [9], 2) modeling implicit taxonomic relation semantics [10], 3) exploiting position-enhanced graph neural network [11], and 4) combining features from multiple sources [12]. In knowledge transfer techniques [9], a model is configured to learn high-dimensional embeddings for the taxonomic concepts and new concepts identified by the text corpus. The possible parent-child relations among the concepts are predicted using semantic features and concepts are added to the taxonomy. In modeling implicit taxonomic relation semantics technique [10], textual taxonomies are expanded by predicting the parents of the new taxonomy concepts. Models built on this technique learn latent representations of the edge semantics with the taxonomy concepts' embeddings for measuring the taxonomic relatedness between concept pairs. In the position-enhanced graph neural network technique (e.g. TaxoExpan), a set of concepts pairs are generated from the existing taxonomy as training data [11]. Using self-supervised data, a model makes predictions about whether a query concept is the direct hyponym (i.e. a word or phrase whose semantic field is more specific than its hypernym) of an anchor concept or not. Last, enrichment systems built by combining features from multiple sources such as Self-supervised Taxonomy Expansion Model (STEAM) exploit natural supervision in the actual taxonomy for enrichment [12]. It extracts distributed features, contextual features, and lexico-syntactic features to provide supplementary information that is important for taxonomy expansion.

The abovementioned techniques of taxonomy enrichment use diverse neural network architectures to apply low-dimensional graph embeddings, deep learning architectures, or graph convolutional networks. The taxonomy enrichment results confirmed by these methods are promising. However, a high degree of technical expertise is required to understand and implement these methods for business organizations. Not only a certain skill set is necessary for their implementation, but also a high-end computing infrastructure needs to be deployed for executing such models. To overcome this technical barrier of technology for business companies, a simplified taxonomy enrichment approach is proposed that is based on exploiting existing Open data sources.

3. Proposed Method

The working of the proposed method (see Fig. 2) of semantic data enrichment is shown by applying it to a current business taxonomy received from a French company FirstECO. The business taxonomy contains around 2,973 concepts related to 18 domains, which are; 1) Public administration, 2) Agriculture, forestry, and fishing, 3) Arts and hobbies, 4) Construction, 5) Trade and distribution, 6) Energy, 7) Education and training, 8) Environment, 9) Finance and insurance, 10) Hospitality, catering and tourism, 11) Extractive industry, 12) Manufacturing industry, 13) Information and communication, 14) IT and telecommunications, 15) Scientific research and development, 16) Human health and social action, 17) Service, and 18) Transport and logistics.

The taxonomy is semantically enriched using relevant concepts got from 4 different sources, which are; 1) lexical datasets, 2) pre-trained word embeddings, 3) linked Open data, and 4) corpus-based vocabulary. The proposed method (see Fig. 2) is presented as a Venn diagram to show the overlapping concepts among 4 data sources that are exploited in this study. However, unique concepts were added during the taxonomy enrichment process. The description of each method of taxonomy enrichment mentioned in Fig. 2 is described below;

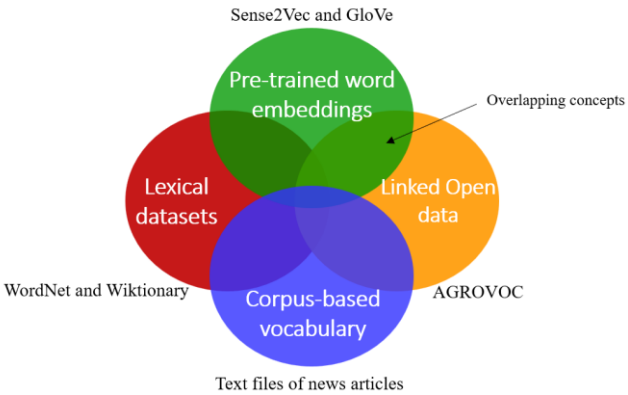

Fig. 2. Proposed method of taxonomy enrichment

a) Lexical datasets

For taxonomy enrichment using lexical data sources (see Table 1), the existence of taxonomy concepts was checked in WordNet [13]. Out of 2973, data of 1044 concepts were found. Their associated data related to synonyms, hyponyms, and definitions were extracted. Part-of-speech tagging is used to tag different words in a concept definition (i.e. a sentence) for identifying a part-of-speech (e.g. nouns, verbs, adjectives, etc). Only nouns and their combinations with adjectives are selected as terms. As existing literature [14] suggests their importance in recognizing lexico-syntactic patterns (e.g. 2-word: {adjective, noun}). To measure the retrieved concepts' (i.e. synonyms, hyponyms, and terms) relevancy to the taxonomy concepts, each taxonomy concept is compared with the extracted concepts using the cosine similarity formula. Cosine similarity is a measure of similarity between two concepts [4-7]. For its calculation in terms of a score interval (-1,1), the concepts are viewed as vectors. The cosine similarity does not depend on the magnitudes of the vectors, but only on their angle [4-9]. Given two vectors of attributes, A and B, the cosine similarity is presented using a dot product and magnitude as;

$$Similarity = \cos(\theta) = \frac{A.B}{\|A\|\|B\|}$$

The cosine similarity score is used to filter the closest matching concepts related to the taxonomy concepts. After the WordNet, Wiktionary [15] is checked against the concepts of a taxonomy. Around 172 concepts were found in the Wiktionary, those were not present in the WordNet and their corresponding data (i.e. synonyms and definitions) was extracted. In total, 1216 taxonomy concepts were found in lexical datasets (WordNet and Wiktionary) giving us a data enrichment of 41%.

Table 1: Statistics of lexical datasets.

No.	Dataset	Nouns	Adjective	Verb
1	WordNet 3.0 [13]	117,798	21,479	11,529
2	Wiktionary [15]	218,629	62,202	58,872

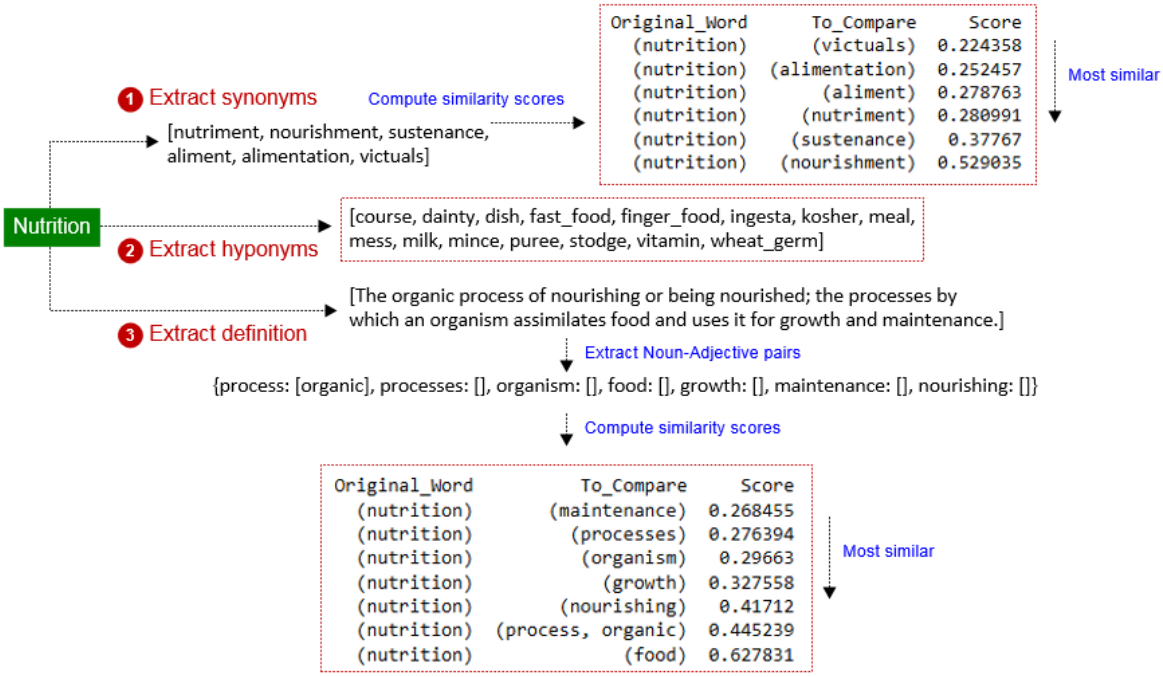

Fig. 3. Extracting synonyms, hyponyms, and a definition of a taxonomy concept

b) Pre-trained word embeddings

After the lexical datasets, pre-trained word embeddings are the most popular choice for taxonomy enrichment in the existing literature [16]. For instance, when solving the Hypernym Discovery problem in SemEval-2018 Task 9, most of the contributors used word embeddings for predicting the likelihood of the relationship between an input word and a candidate [4]. In the literature [17], word embedding models such as Word2Vec, Sense2Vec, GloVe, and FastText are present. For this study, pre-trained word embedding models Sense2Vec [18] and GloVe [19] are used. The reason for selecting them was entirely based on the extent of word vectors` coverage and size. Around 355 concepts out of 1,177 (achieved data enrichment in total: 12%), which were not found in the lexical datasets were present in the Sense2Vec and GloVe data extracts, and their corresponding data (see Fig. 4) was retrieved, and later added to the business taxonomy.

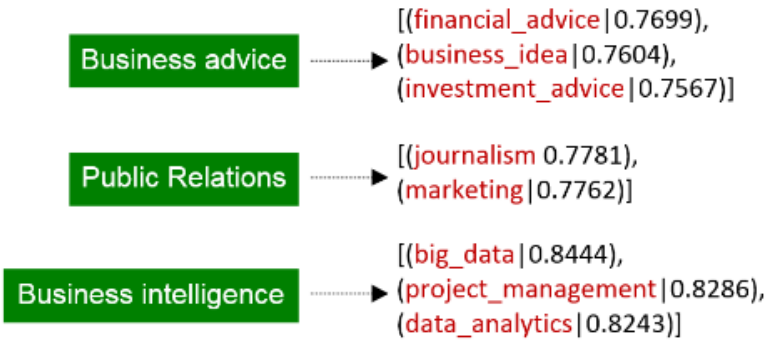

Fig. 4. Word embeddings with similarity scores

c) Linked Open data

Based on the taxonomy domains (as discussed above), AGROVOC [20] is selected for linked data coverage. It is the largest linked Open dataset related to food, nutrition, agriculture, and the environment concepts. It is based on semantic web technologies to link the concepts with other knowledge organization systems. AGROVOC is a Resource Description Framework (RDF) and Simple Knowledge Organization System eXtension for Labels (SKOS-XL) concept scheme [20]. In the AGROVOC, the classical thesauri relations are expressed by the SKOS predicates skos: broader and skos: narrower (see Fig. 5). Data of 874 taxonomy concepts that were not found in the lexical and pre-trained word embedding models was retrieved (i.e. data enrichment: 29%) from the AGROVOC using its SPARQL Protocol and RDF Query Language service.

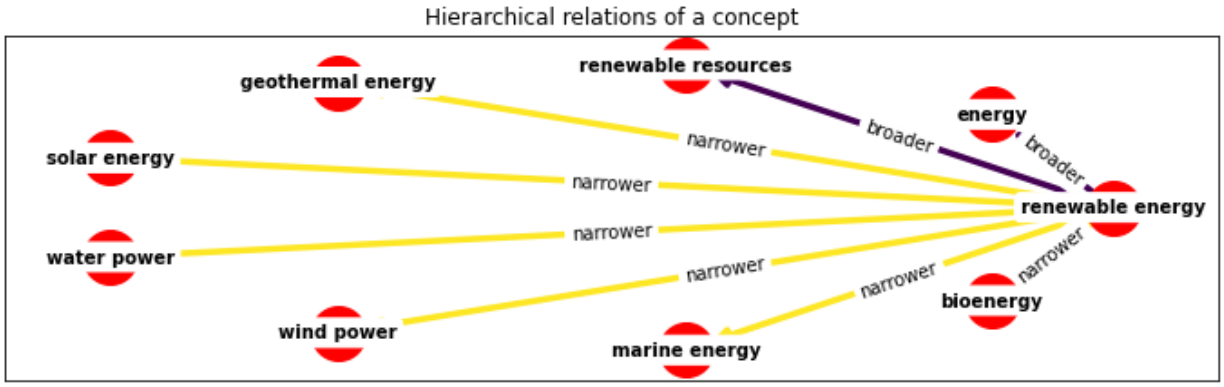

Fig. 5. Concept data from AGROVOC

d) Corpus-based vocabulary

The aim of taxonomy enrichment through corpus-based technique comprises finding suitable concepts which are related to existing concepts of a given business taxonomy. To get the unique thematic concepts from online news documents (i.e. as text files), a corpus of 26,503 text files carrying 2,698,339 words is used. The news articles were extracted from multiple online news sources during the period of 2017 to 2021. Basic pre-processing steps such as stop-word removal, and filtering of special characters, hashtags, web links, numbers, email addresses, and punctuations are executed. Then, lemmatization (i.e. returns the base or dictionary form of a word) is performed to lemmatize the words. The given business taxonomy is only based on thematic concepts. Irrelevant data such as spatiotemporal (i.e. location and time) and numeric (e.g. cardinal, ordinal, and percentage) entities are identified (287,885 found in total) using the named Named-entity recognition (NER) technique of NLP [21] (see Fig. 6) and removed from the corpus. To end, based the occurrence counts of various concepts in a corpus of text files, a vocabulary of 52,439 concepts (see Fig. 7) is got after the pre-processing and duplications' removal from the news documents.

The vectors of concepts from the taxonomy as well as from the vocabulary are extracted using the pre-trained word vectors' dataset (i.e. GloVe-wiki-gigaword-300 trained on Wikipedia (2014) and the Gigaword corpus, i.e. a large collection of newswire text) which is prepared using the GloVe. Here, the taxonomy enrichment is treated as a ranking task in which the correct possibilities of concepts based on cosine similarity scores should be at the top of a candidate list. The concepts 267 in total having the cosine similarity scores greater than 0.6 are selected for the taxonomy enrichment. Whereas, this similarity threshold of detecting the relevant concepts can be changed as per the user requirements.

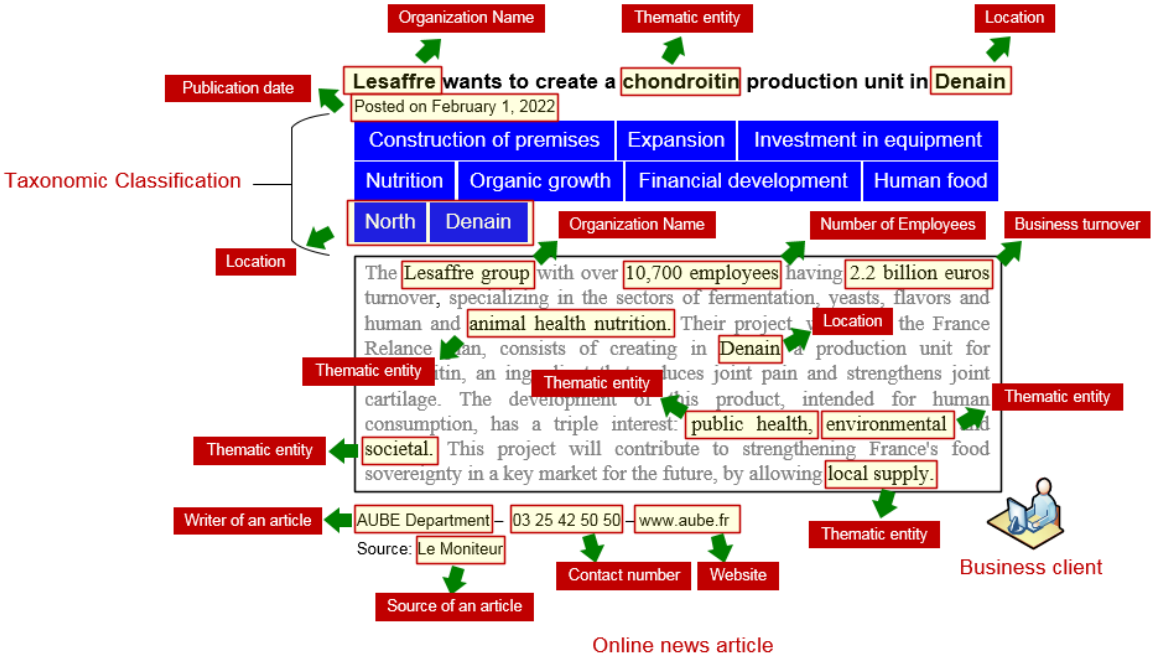

Fig. 6. Recognition of entities using NER

Fig. 7. Extracted features from a corpus of news articles

4. Evaluation and Discussion

The proposed method of semantic data enrichment uses 4 different sources, including lexical datasets, pre-trained word embeddings, linked Open dataset, and a corpus-based vocabulary. The selection of these sources was made to exploit publicly available data sources with minimum computing and labor costs for a business company. For the lexical data, the Natural Language Toolkit (NLTK) Python library [22] is used to extract data from WordNet and a WiktionaryParser [15] is used for Wiktionary's data retrieval. In addition, pre-trained word embedding vectors for Sense2Vec [18] and GloVe [19] are downloaded from their official GitHub repositories and were later linked with the Python program. To fetch the linked data through AGROVOC, a SPARQLWrapper (i.e. a wrapper for a remote SPARQL endpoint) is used [20], i.e. available, on its official website https://agrovoc.fao.org/sparql. Identification of relevant texts as synonyms, hyponyms, definitions (i.e. in noun-adjective pairs), contextually keyed words representation from pre-trained word embeddings, narrower concepts, and the most similar concepts recognized

from the corpus of text articles is achieved to enrich original concepts of a business taxonomy. For instance, 'renewable energy', a business concept, is enriched with multifaceted data sources (see Table 2).

Table 2. Enrichment of a taxonomy concept

Business concept	**Semantic enrichment using multifaceted data sources**			
	WordNet/ Wiktionary	Sense2Vec/ GloVe	AGROVOC	Corpus-based vocabulary
Renewable energy	Hydroelectric, hydropower solar energy, solar power water power, wind energy wind power	Biofuels, biomass renewables, clean energy green energy, sustainable energy fossil fuels	Bioenergy, geothermal energy, marine energy	Renewable energy sources, Alternative energy

During the insertion of new concepts into a business taxonomy, the identified new concepts, as well as taxonomy concepts, are converted into vector representations, and later a cosine similarity index is used to identify semantic similarities [23] between them. To evaluate the relevancy of added concepts against the taxonomy concepts, the unwanted concepts were removed manually from the list. The reason for opting for a manual method of taxonomy quality evaluation is because of the small size of the company taxonomy. Whereas, the manual method is not recommended for bigger taxonomies with thousands of concepts. During the evaluation, it is found that even a cosine similarity matrix with a score of 0.6 (as discussed above in the previous section) is used to select only the top relevant concepts. But, there are still irrelevant concepts tagged. For instance, a taxonomy concept i.e. Bus shelter is tagged with irrelevant concepts having a high similarity score highlighted in red in Table 3 below.

Table 3. Tagging of relevant and irrelevant concepts

Taxonomy concept	Enriched concepts with cosine similarity score
Bus shelter	(bus_stop, 0.8006), (car_park, 0.7856), (carpark, 0.7749), (parking_garage, 0.7748), (sidewalk, 0.7713), (front_steps, 0.7569), (front_window, 0.7561), (balcony, 0.7554), (bus_station, 0.7554), (park_bench, 0.7554)

After removing the irrelevant concepts, the taxonomy enrichment achieved based on the proposed method is around 91%. Added concepts will contribute to enhancing the working of the keyword-based search process that is used for the classification of business articles. The news articles are classified as relevant to a certain taxonomy concept if their added corresponding concepts (see Table 2) are found in the text. In addition, data of 267 taxonomy concepts was not found in selected data sources. A few examples of these concepts are given in Table 4.

Table 4. Out-of-vocabulary taxonomy concepts

No.	Taxonomy concepts	No.	Taxonomy Concepts	No.	Taxonomy Concepts
1	Prefabricated wooden building	7	Hydroelectric power plant	13	Short-term car rental
2	Ready-mixed concrete	8	Ceramics for agricultural use	14	Musical instrument rental
3	Plastic boiler making	9	Finance and insurance	15	Intellectual property leasing
4	Leather dressing and tanning	10	Theater management	16	Electrical lighting equipment
5	Ceramic article for domestic use	11	Lubricating petroleum oil	17	Lifting and handling equipment
6	Automotive equipment trade	12	Botanical and zoological garden	18	Medical and surgical equipment

This problem arises from certain concepts that appear frequently, are derived from the combination of different words, or resulted from spelling errors. In our case, the data related to these concepts were not found in WordNet and Wiktionary, and they were not part of the training text vocabulary of the (e.g. Sense2Vec and GloVe) models which are exploited in this study. These words that are not part of the training text vocabulary are called Out-of-Vocabulary (OOV) words [24]. The OOV problem is very common in NLP. These OOV words need to be properly handled to avoid degrading the data quality in the NLP processing applications. So, to deal with this problem of

OOV words, different solutions leveraging various techniques of machine learning and artificial intelligence are present in the literature [23]. One of the existing solutions is to train a neural network to break the words into several sub-words and then feed them to the neural network for relevancy detection.

5. Conclusion

In this paper, a method based on combining data from different datasets comprising lexical datasets, pre-trained word embeddings, linked Open data, and corpus-based thematic features extraction from news articles is presented. Around 91% of taxonomy concepts were found in selected datasets and their associated semantic data was extracted for taxonomy enrichment. The business taxonomy enriched by this method is focused, and its depth can be influenced by varying the cosine similarity threshold in selecting relevant concepts. A high threshold creates a more focused taxonomy. Whereas, a low similarity threshold may encompass the extraction of irrelevant concepts. The presented method is simple and low cost, which enables organizations to transform generic taxonomies into specialized taxonomies as per their line of business through Open data sources. The main benefit of this method is speed. Given Open datasets and a huge corpus of text articles, a fully-fledged taxonomy is semantically enriched in minutes. Another benefit achieved is that the enriched taxonomy focuses on what actually appears in the news documents. Only top relevant concepts are extracted from the news documents based on the occurrence counts rather than background information. Additionally, the proposed taxonomy enrichment method is applicable to other domains such as life sciences, healthcare, environmental sciences, etc. Future work could be to examine the performance of the enriched taxonomy for news articles' classifications. Apart from the benefits of the proposed method, there exist limitations as well. The limitation of the work is that the n-grams (i.e. continuous sequence of words e.g. Automotive equipment trade) were not treated wholly for finding the related concepts to enrich a taxonomy. The n-grams were tokenized, and each token was searched in data sources. However, this approach is not preferred as it does not keep the original sequence of n-grams. Consequently, the combination of tokens changes the context of concepts found against the n-grams of a taxonomy. Moreover, in the case of concept retrieval from pre-trained embeddings, the training of data plays an important role in training the model for diverse business use cases. If an n-gram of taxonomy is input into the model where the n-gram is not recognized in training, the word embeddings generated will be of poor quality, and the representation of the n-gram meaning might not be accurate as well. To avoid such scenarios, a larger training corpus text is required.

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Response to Reviewers;

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	A	B	C	D
1		**Reviewer 1**		
2	No.	**Comments**	**Status**	**Response**
3	1	The authors propose an approach for semantic enrichment that leverages lexical datasets, pre-trained word embedding, linked open data and a corpus-based vocabulary. Concepts are enriched by factoring in their respective concept similarity across the varying sources. Although the topic is relevant and the methods related to it are adequate the contribution is marginal. The article does lack detail for the **reproduction of the work** and more discussion should be placed on **metrics**, along with an appropriate ablation study that looks at varying parameters.	Done	A paragraph is added to describe how the proposed method can be reproduced to achieve the taxonomy enrichment in the **Evaluation and Discussion** section. Whereas, the variation of the cosine similarity metric is discussed in the **Conclusion.**
4				
5		**Reviewer 2**		
6		**Comments**	**Status**	**Response**
7	1	Abstract can be misleading. From reading the abstract, I first misunderstood the paper. The term taxonomy is used in different ways. From the abstract I expected the paper to propose some sort of framework of general definition and terminology. However it proposes an approach to create a taxonomy on the NLP sense to work on the problem of news article classification. It became clear from the introduction section, which started quite differently from what was said in the abstract. **I suggest to revise the abstract.**	Done	The abstract is revised.
8	2	Evaluation should have more depth The evaluation of the overall method is not very detailled. It is shown in the section Discussion. I suggest to rename the section to Evaluation and Discussion and **enhance the evaluation part.** It is currently shown with the number of concepts that were not found and a small list of examples. That should be evaluated and discussed in more depth.	Done	The section is renamed as per the comments. Also, a list of examples is extended by adding 12 new concepts which were not found. Also, the explanation is added and highlighted in red.
9	3	**General Limitations** of the approach are not discussed	Done	A paragraph is added in the **Conclusion** to describe the method limitations.
10	4	Fig.2 is stated to show the method, however, it actually shows the overlap of the data sources. I suggest to enhance the figure, making clear that the method uses the overlap.	Done	The figure is enhanced.
11	5	Minor issues: (e.g. 2-word: adjective, noun}) => } bracket the retrieved concepts (i.e. synonyms, hyponyms, and terms) relevancy => retrieved concepts' ... cosine similarity: A.B => this operator is not common for the dot product, I suggest to rather use the vertically centered one	Done	The letter corrections are made. The cosine similarity formula is updated as per its reference article.